

**The Short-Term Impact of Global Regulatory News from Major Economies on Bitcoin
and Ethereum Price Volatility**

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Abstract

This study investigates the short-term impact of regulatory announcements from the United States, China, and the European Union on the price behavior of Bitcoin (BTC) and Ethereum (ETH). Using an event study methodology, the research quantifies abnormal returns and volatility within a one-day window following key policy announcements between January 2023 and April 2025. The study categorizes regulations into five types—Bans, Taxation Policies, KYC/AML Regulations, Securities Classifications, and CBDC Initiatives—to analyze their distinct market effects. Findings show that Ethereum is significantly more sensitive to regulatory news than Bitcoin, particularly in response to enforcement-heavy categories such as bans and securities designations. China's regulatory actions triggered the most negative reactions, while the European Union generated more stable responses. The results highlight critical differences in investor sentiment across jurisdictions and asset types, offering actionable insights for traders, policymakers, analysts, and financial modelers. The study contributes to regulatory risk assessment in digital asset markets and emphasizes the need for jurisdiction-aware trading strategies and policymaking.

Keywords

Cryptocurrency regulation, price volatility, Bitcoin, Ethereum, event study methodology, abnormal returns, comparative analysis, digital assets, financial policy impact, short-term trading dynamics

Introduction

Background of the Study

Cryptocurrencies like Bitcoin (BTC) and Ethereum (ETH) have transformed from speculative digital tokens to internationally recognized financial assets during the last ten years, now comprising over 60% of the total cryptocurrency market capitalization (Katsiampa, 2019; Nakagawa & Sakemoto, 2022). Their independence from conventional banking institutions makes them appealing as decentralized instruments. But because of this precise characteristic, they are also more susceptible to outside shocks, particularly regulatory announcements (Auer & Claessens, 2018; Shanaev et al., 2020). The intrinsic volatility of both Bitcoin and Ethereum, which is frequently many times greater than that of conventional fiat currencies, makes them perfect subjects for studies of short-term market reactions (Baur & Dimpfl, 2018; Brini & Lenz, 2024).

Even studies have long been used in traditional financial markets to evaluate how external events, like policy changes or earnings announcements, affect asset prices (Timmermann et al., 2023). According to recent research, this methodology is also becoming more popular in cryptocurrency studies, especially when it comes to assessing unusual returns and volatility spikes that coincide with regulatory events (Chokor & Alfieri, 2021; Lyócsa et al., 2020). Despite this progress, several research gaps remain: (1) investor sentiment is often mentioned but rarely quantified using volatility models, (2) most studies generalize across a broad set of crypto tokens instead of focusing on market leaders like BTC and ETH, and (3) regulatory effects are usually studied in isolation per country, overlooking the opportunity for comparative insights (Bonaparte & Bernile, 2023; Saggu et al., 2023).

The rise of cryptocurrencies, especially Bitcoin (BTC), has changed global financial markets by incorporating decentralized, borderless, and generally unregulated assets into mainstream investment portfolios. Since their introduction, these digital currencies have

experienced significant price volatility, driven by speculative trading, market sentiment, and global economic concerns (Corbet et al., 2019). However, a growing body of research indicates that regulatory actions have an important role in shaping cryptocurrency market dynamics. For instance, regulatory notifications can cause quick changes in trade volume and price movements (Foley et al., 2019).

Prior studies (Auer & Claessens, 2018; Chokor & Alfieri, 2021; Shanaev et. al, 2020) have looked at the impact of various policy announcements, but the findings vary depending on the type, jurisdiction, and severity of the regulatory action (Li, 2021). In general, these studies showed that regulatory announcements cause negative reactions and are perceived in a negative light. Both the US and China have fragmented financial regulatory structure. Multiple regulators in each country monitor the cryptocurrency market depending on their jurisdiction in order to address distinct concerns. For instance, in the US, the Commodity Futures Trading Commission regulates cryptocurrencies like commodities, including their derivatives markets (Li, 2021). Similarly, China has many agencies in control of various parts of financial markets, institutions, and associated risks. For instance, the PBOC, China Securities Regulatory commission (CSRC), and China Banking and Insurance Regulatory Commission (CBIRC) are in charge of overseeing macroprudential risks in banking, securities and insurance, respectively (Li, 2021).

The years 2023-2025 represent a watershed moment in cryptocurrency governance, marked by heightened global attention following the collapse of significant platforms such as FTX in late 2022 (Coelho et al., 2024). This has resulted in a wave of legislation proposals and enforcement activities across the globe, all aimed at restoring investor confidence and preventing systemic risk (Shine, 2024). At the same time, media amplification of these events can influence public mood and drive asset markets to react even before formal regulation is implemented (Zhang et al., 2023). The lack of regulatory uniformity further

adds uncertainty, as countries differ substantially in their definitions and rules of digital assets-some consider them securities, others commodities, and others outright prohibit them (OSL Research Team, 2025). This fragmented regulatory landscape necessitates comparative examination. Understanding how investors react differently to regulatory news in the US, EU, and China can reveal crucial jurisdiction-specific dynamics and inform the establishment of more globally coherent cryptocurrency laws (Frediani, 2024).

Problem Statement

Despite the growing popularity of cryptocurrencies and their benefits, particularly for countries and investors leveraging their decentralized nature, there remains a lack of structured research on regulatory sentiment and its immediate effects on the market (Saggu et al., 2023). Existing studies often mention emotional responses to regulatory events but rarely quantify these reactions using short-term return or volatility models (Bonaparte & Bernile, 2023). Although some research has examined how cryptocurrency markets respond to regulatory news, few studies have focused on isolating the immediate, short-term (within one day) impact of such announcements across major global economies (Bains et al., 2022; Li & Sugimoto, 2023). Moreover, while Bitcoin (BTC) and Ethereum (ETH) dominate the cryptocurrency market, prior research frequently generalizes findings across a broad set of tokens or altcoins. This study addresses these gaps by focusing exclusively on BTC and ETH within a one-day event window, aiming to deliver a more precise and actionable understanding of how different types of regulatory interventions affect leading cryptocurrencies in the short term.

Objective of the Study

The research aims to accomplish a series of focused objectives that guide the analysis of regulatory announcements and their short-term impact on the cryptocurrency market. Each objective reflects a core step in the overall methodology of the study.

1. To identify and classify significant cryptocurrency regulatory announcements from the United States, European Union, and China between 2023 and 2025
2. To collect and preprocess high-quality daily price and volume data for Bitcoin and Ethereum surrounding each regulatory event
3. To measure abnormal returns and price volatility of BTC and ETH within a short-term event window following each regulatory announcement
4. To evaluate the comparative effects of different regulatory announcement types on BTC and ETH market responses
5. To compare the differential impact of regulatory actions across jurisdictions (USA, EU, China) and between Bitcoin and Ethereum

Scope and Limitations

The research will examine official regulatory announcements related to cryptocurrency that originate from the United States, China, and the European Union. It will focus on the short-term effects of these announcements on Bitcoin and Ethereum using a one-day event window from the date of the announcement to the following day. The regulatory categories include Bans, Taxation Policies, Know Your Customer (KYC) and Anti-Money Laundering (AML) rules, Securities Classifications, and Central Bank Digital Currency (CBDC) Initiatives. The study will cover the period from January 1, 2023, to April 30, 2025. This timeframe was chosen due to the increased global focus on cryptocurrency regulation during these years (CPI, 2024; Shine, 2024). Data will be gathered using automated scraping tools and application programming interfaces (APIs) to ensure efficiency and accuracy.

The research will be restricted to formal, publicly released announcements, disregarding unofficial statements or casual remarks. The analysis will focus only on

short-term market reactions and will not explore long-term policy effects or delayed market responses. Sentiment analysis and indirect metrics will also be excluded in the dataset.

Significance of the Study

The findings of this study will provide valuable insights into how regulatory announcements from major global economies, specifically the United States of America, China, and the European Union, on Bitcoin and Ethereum price volatility and unexpected price movements. The results will be significant for the following groups:

Cryptocurrency Investors and Traders

The study will help investors and traders understand how regulatory changes from these major economies impact Bitcoin and Ethereum, improving decision-making during regulatory events.

Regulatory Bodies and Policymakers

The findings will assist regulatory bodies in assessing the immediate effects of their announcements on cryptocurrency markets, helping refine regulatory strategies.

Financial Analyst and Economist

The findings will aid analysts in forecasting market trends and understanding the influence of regulatory news on the digital asset market.

Data Analysts

The study offers data analysts a structured framework for quantifying the market impact of regulatory events, enabling them to build more accurate forecasting models and real-time monitoring tools.

Future Researchers

This study will serve as a reference for future research on the short-term impact of regulatory events on the cryptocurrency market.

Methods

This study employs a quantitative event study methodology to assess the short-term effects of regulatory announcements on the cryptocurrency market, specifically focusing on Bitcoin (BTC) and Ethereum (ETH). The approach unfolds in three key phases: first, the systematic identification and categorization of major regulatory announcements from official government and institutional sources; second, the collection and preprocessing of high-frequency daily price and volume data for BTC and ETH; and third, the statistical analysis of price behavior within a Day 0 to Day +1 event window. This includes computing abnormal returns, analyzing volatility shifts, and testing for significance using descriptive statistics, t-tests, and GARCH-based models. A robustness check employing an extended [-1, +1) window will be conducted to confirm the reliability of observed effects.

The selection of this methodology is supported by multiple studies that have explored how crypto markets respond to external policy shocks. Bitcoin's volatility has been found to increase sharply following regulatory announcements, affirming the value of event studies in capturing market responses (Lyócsa et al., 2020). Statistically significant abnormal returns have also been observed around regulatory events, suggesting that crypto investors react swiftly to policy news (Chokor and Alfieri, 2021). Additionally, due to the high volatility and clustering effects in crypto markets, traditional parametric models should be adapted or replaced with non-parametric methods for better accuracy (Amsden, 2019).

Broader financial literature further supports the timeliness and efficiency of market reactions to news. Equity markets have been shown to adjust within milliseconds following earnings announcements, while near-instantaneous price movements have been observed in foreign exchange markets after macroeconomic disclosures (Christensen et al., 2023). Therefore, these studies indicate that even a one-day event window can effectively capture meaningful market responses.

Regulatory Event Identification and Classification

This study categorizes regulatory announcements into five primary categories: Bans, Taxation, Policies, Know Your Customer/Anti-Laundering (KYC/AML) Regulations, Securities Classifications, and Central Bank Digital Currency (CBDC) Initiatives. These categories were selected due to their distinct legal implications, differentiated effects on market activity, and frequent references in financial regulation literature and global policy discussions. Classifying events in this way enables the study to systematically evaluate how each type of intervention affects short-term market behavior and asset valuation.

To achieve systematic categorization, a rule-based approach is employed. This approach uses a set of predefined keywords that align with each of the five categories. We can automatically classify regulatory events based on their content by applying keyword-based filtering. Below are the categories and how they are identified using this methodology:

Bans represent the most restrictive category, typically involving a complete prohibition of cryptocurrency trading, mining operations, or access to exchanges. Such announcements often result in immediate and pronounced market disruptions, including sharp increases in volatility and investor sell-offs. Prior students have demonstrated that bans, particularly in jurisdiction such as China can significantly destabilize the market and prompt capital outflows (Auer & Claessens, 2018; Shine, 2022).

Taxation Policies are comparatively moderate regulatory measures aimed at enhancing transparency and compliance without prohibiting cryptocurrency use. They reshape incentives by affecting holding periods, transaction frequency, and overall compliance with national reporting standards (EY Global, 2023; OECD, 2020)

KYC/AML Regulations do not restrict usage directly but impose due diligence requirements to prevent illicit transactions. They are focused on improving the transparency

and traceability of digital asset flows, especially through exchanges and service providers. Literature from both enforcement bodies and analytics firms shows that these frameworks are critical for curbing crypto-related crime while legitimizing broader market participation (FATF, 2021; Chainalysis, 2022; U.S. Treasury, 2022).

Securities Classifications determine whether crypto assets are subject to securities law, impacting how they are offered, traded, and disclosed. These designations impose regulatory burdens but do not outright prohibit trading. As shown in various U.S. SEC enforcement cases, such classification can trigger changes in market access and legal treatment of tokenized assets (U.S. SEC, 2023; Hinman, 2018; World Economic Forum, 2020).

CBDC Initiatives are a different kind of regulatory force. Instead of restricting cryptocurrencies, they introduce government-backed digital currencies that may influence public trust and investor demand. These state-sponsored alternatives could reshape expectations around decentralization, monetary control, and digital finance infrastructure (BIS, 2023).

The rule-based categorization methodology begins by extracting regulatory events from RSS feeds and web scraping, which provide real-time data on crypto-related regulatory announcements from authoritative sources, government websites, and financial media outlets. The extracted data is then passed through a rule-based system that utilizes keyword matching to classify each event into one of five categories: Bans, Taxation Policies, KYC/AML Regulations, Securities Classifications, or CBDC Initiatives. To achieve this, each event's content is filtered using predefined keyword lists corresponding to each category, with the event being checked for keywords like "prohibited," "taxation," or "compliance" to determine the appropriate classification. Once the keywords are identified, the event is assigned to the relevant category. After classification, the study systematically evaluates how each regulatory

intervention impacts short-term market behavior and asset valuation, enabling a comparative analysis across the U.S., EU, and China.

Overall, this classification framework captures the most relevant and frequently occurring types of crypto-related regulation across the U.S., EU, and China—making it both practical and theoretically grounded for comparative analysis.

Cryptocurrency Market Data Collection and Preprocessing

This phase involves acquiring historical daily market data for Bitcoin (BTC) and Ethereum (ETH) and preparing it for analysis. Data will include closing prices, trading volume, and volatility indicators from January 1, 2023, to April 30, 2025. Primary sources will be CryptoCompare and CoinGecko APIs, both of which offer reliable and high-frequency access to digital asset pricing.

The collected data will be cleaned for consistency, standardizing timestamps, handling missing values, and removing duplicates or outliers. It will then be aligned with regulatory event dates to match BTC and ETH market behavior.

Logarithmic returns will be computed to standardize data for statistical tests and volatility models (Amsden, 2019). Using APIs ensures efficiency, consistency, and reduces manual errors during data collection and updating (Lyócsa et al., 2020).

Event Window and Abnormal Return Computation

This section focuses on calculating abnormal returns for BTC and ETH surrounding regulatory announcements. The main event window is Day 0 (announcement date) to Day +1. A robustness check will use a $[-1, +1)$ window to capture possible early or delayed reactions.

Cumulative Abnormal Return (CAR) will be computed for both event windows. To test significance, one-sample t-tests will be applied to AR and CAR values. GARCH (1,1) models will be used to assess volatility before and after the announcements (Amsden, 2019; Chokor and Alfieri, 2021).

This approach is grounded in financial event study literature, where 1–3 day windows are typical for isolating market responses to impactful news (Lyócsa et al., 2020; Malkiel, 2003).

Impact Analysis Based on Regulatory Announcement Types

To evaluate whether different types of regulatory announcements trigger varying levels of market response, each identified event will be categorized into one of the following types: Ban, Taxation Policy, KYC/AML Regulation, Securities Classification, or CBDC Initiative. These categories reflect common regulatory mechanisms observed globally and are derived from both academic literature and policy discourse (Chokor and Alfieri, 2021; Amsden, 2019).

For each category, cumulative abnormal returns (CARs), trading volume changes, and volatility shifts will be computed and compared. Statistical comparisons will be conducted using Kruskal-Wallis tests. These tests will help determine whether the type of regulation has a statistically significant effect on the magnitude or direction of the market reaction.

This analysis enables the differentiation of investor sensitivity across regulatory mechanisms and provides insight into which types of announcements are perceived as most disruptive or stabilizing. Prior research suggests that enforcement-heavy announcements, such as bans or securities classifications, tend to provoke stronger market reactions than more procedural rules like KYC or tax policies (Lyócsa et al., 2020).

Jurisdictional and Asset-Specific Comparative Analysis

This final stage examines how the impact of regulatory announcements varies by jurisdiction (United States, European Union, China) and by cryptocurrency (BTC vs. ETH). For jurisdictional analysis, regulatory events will be grouped by country of origin. Mean CARs and volatility changes will be compared across jurisdictions using Mann–Whitney U tests to assess differences in market sensitivity.

For asset-specific analysis, the responses of BTC and ETH to the same event types will be compared directly. This helps identify whether one cryptocurrency is more reactive or resilient to specific policy signals. Such comparative analysis is particularly important given differences in use cases, market structures, and investor bases between Bitcoin and Ethereum (Amsden, 2019; Lyócsa et al., 2020).

By assessing both jurisdictional and asset-specific reactions, the study aims to provide a multidimensional understanding of how regulatory news influences crypto markets across different contexts.

Results and Discussion of results

Regulatory Announcements Classification

The process of classifying regulatory announcements began with web scraping, where the goal was to gather regulatory event data from various sources, particularly through RSS feeds. Using a Python-based script, we collected relevant RSS feed entries related to cryptocurrency regulations from key government bodies and media outlets in the U.S., the EU, and China.

```
2025-05-10 17:56:13,910 - INFO - Scraping complete. Saved 352 entries to ../regulatory_announcement.json.
```

Figure 1: *Collected Regulatory Entries from RSS Feeds*

Figure 1 shows that we successfully collected 352 entries, which were stored in a structured format for further analysis. Each entry includes key information such as the title, publication date, description, keywords, and source, information essential for subsequent classification. The next step was to apply a keyword-based classification approach, where these entries were labeled based on the presence of specific keywords.

```
{
  "title": "SEC Announces Agenda, Panelists for Roundtable on Tokenization Plus Date Change for Roundtable on DeFi",
  "link": "https://www.sec.gov/newsroom/press-releases/2025-72",
  "published": "Mon, 05 May 2025 09:00:00 -0400",
  "description": {
    "description": "The Securities and Exchange Commission's Crypto Task Force has announced the agenda and panelists for its May 12 roundtable",
    "category": "Crypto Regulation",
    "keywords": [
      "crypto",
      "defi",
      "exchange",
      "securities",
      "securities and exchange commission",
      "token"
    ]
  },
  "country": "United States",
  "entry_number": 1
},
```

Figure 2: *Sample Structured RSS Entry Output*

Figure 2 illustrates the JSON output format of the scrapped RSS feed entries. Each RSS feed entry contains structured information such as the article title, description, category, country, and publication date. These entries were parsed using Python’s *feedparser* library, allowing consistent and efficient data extraction. After collecting the data, we moved on to the classification stage.

The classification was performed using a keyword-based matching approach. Predefined keyword sets were created for each of the five regulatory categories defined in Section 2.1: *Bans*, *Taxation Policies*, *KYC/AML Regulations*, *Securities Classifications*, and *CBDC Initiatives*. These keywords were selected based on terminology commonly used in regulatory literature, official government statements, and financial media coverage. For instance, terms such as “ban”, “prohibit” or “illegal” were indicative of Bans, while keywords like “fiscal”, “tax” or “levy” were associated with Taxation Policies.

entry_number	29
title	Federal Reserve Board announces approval of proposal by EverBank Financial Corp
published	Fri, 14 Mar 2025 18:30:00 GMT
country	United States
description	Federal Reserve Board announces approval of proposal by EverBank Financial Corp
keywords	['approval', 'ban', 'federal reserve', 'rba']
link	https://www.federalreserve.gov/newsevents/pressreleases/orders20250314b.htm
label	Bans

Figure 3: *Keyword-Based Classification Output of Regulatory Entries*

Figure 3 illustrates that each RRS entry was scanned for the presence of these predefined keywords. When relevant terms appeared in either the title, keywords, or description, the corresponding regulatory categories were assigned to the entry.

Once the entries were accurately categorized, they became suitable for use in the next phase of the study, which involves a comparative analysis across jurisdictions (U.S., EU, and China) to examine how different types of regulatory interventions influence short-term market behavior and asset valuation.

Cryptocurrency Market

Historical price data for Bitcoin (BTC) and Ethereum (ETH) was obtained using the CoinDesk Market Data API, which offers institutional-grade cryptocurrency market indices through its `/index/cc/v1/historical/days` endpoint. This endpoint was selected because it provides volume-weighted average daily prices, adjusted for time and outlier bias via CoinDesk's `cadli` index family—ensuring a reliable, standardized benchmark that reflects 24-hour global trading behavior. The dataset spans from January 1, 2023, to April 30, 2025, aligning precisely with the timeframe under study for regulatory announcement impacts.

The API call was parameterized with careful consideration for data quality and suitability for event study analysis. First, the market parameter was set to `"cadli"` to leverage CoinDesk's core index methodology, which combines multiple exchange data feeds while penalizing stale pricing and abnormal spikes. This minimizes the influence of low-volume or manipulated markets. Second, the instrument parameter was specified separately for `"BTC-USD"` and `"ETH-USD"` to capture dollar-denominated valuations of the two most capitalized and actively traded cryptocurrencies.

To maintain the granularity needed for precise return computation, the aggregate parameter was set to 1, ensuring that data was returned at a daily frequency with no interval smoothing. The fill parameter was set to true to ensure the continuity of the time series even

during days of low trading activity, avoiding data gaps that would distort return calculations. The `apply_mapping` parameter was also set to `true`, enabling the API to internally resolve any symbol variations across exchanges and return harmonized asset identifiers. Lastly, the `response_format` was set to "CSV" for easier preprocessing and integration into the analysis pipeline.

Computation of Abnormal Returns and Cumulative Abnormal Returns

To assess the short-term market reaction to regulatory announcements, this study employed an event study methodology focused on the calculation of abnormal returns (AR) and cumulative abnormal returns (CAR) for Bitcoin (BTC) and Ethereum (ETH). Abnormal returns measure the deviation of actual observed returns from what would have been expected in the absence of the event. This isolates the effect of each regulatory announcement from normal market behavior.

First, daily **logarithmic returns** were computed for both BTC and ETH:

```
Python
btc_df["RETURN"] = np.log(btc_df["btc_close"] / btc_df["btc_open"])
eth_df["RETURN"] = np.log(eth_df["eth_close"] / eth_df["eth_open"])
```

To determine the expected return in the absence of a regulatory shock, a *benchmark return* was calculated for each asset using the average return on **Day -1** across all events.

```
Python
btc_mean_return = btc_df[btc_df["day_relative"] == -1]["RETURN"].mean()
eth_mean_return = eth_df[eth_df["day_relative"] == -1]["RETURN"].mean()
```

This served as a proxy for normal price behavior immediately prior to the announcement window, in line with prior event study literature where estimation windows are either uniform or conditionally constrained by data availability.

The abnormal return (AR) for each observation in the event window was then computed as:

```
Python
btc_df["AR"]=btc_df["RETURN"] - btc_mean_return

eth_df["AR"]=eth_df["RETURN"] - eth_mean_return
```

For each regulatory event, **Cumulative Abnormal Returns (CAR)** were computed by summing the ARs over the main event window [Day -1, Day 0, Day +1):

```
Python
btc_car= btc_df[btc_df["day_relative"].isin([-1,0,
1]))).groupby("entry_number")["AR"].sum()

eth_car= eth_df[eth_df["day_relative"].isin([-1,0,
1]))).groupby("entry_number")["AR"].sum()
```

CAR serves as the primary metric for measuring the overall short-term market reaction to a regulatory event. Each CAR value was tagged with the corresponding event metadata, including country of origin and regulatory classification (e.g., Ban, Taxation, KYC/AML), enabling further group-wise analysis.

```
Python
.agg({
    "AR": "sum",
    "country": "first",
    "label": "first"
})
```

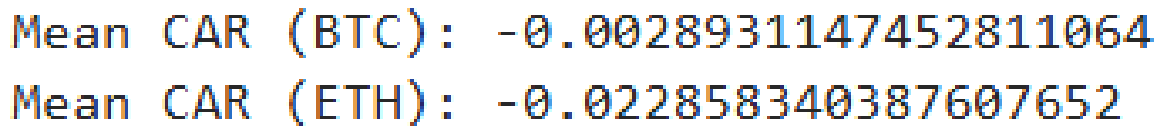
Impact of Regulatory Announcements

Average Market Response to Regulatory Events

The mean CAR shown in Figure 4 reflects the average direction and size of market response to all regulatory announcements. This serves as a foundation for determining if the overall attitude is positive or negative.

- **Bitcoin (BTC):** Mean CAR = **-0.0029**
- **Ethereum (ETH):** Mean CAR = **-0.0229**

Ethereum has a stronger average negative reaction than Bitcoin, showing a higher sensitivity to regulatory changes or developments.



Mean CAR (BTC): -0.0028931147452811064
Mean CAR (ETH): -0.022858340387607652

Figure 4: *Mean CAR*

Both Bitcoin and Ethereum typically see price drops in the immediate wake of international regulatory announcements, according to the negative mean CAR values. With a value of -0.0229, Ethereum is much more sensitive to regulation in the short term than Bitcoin, with a value of -0.0029.

Given that Ethereum serves as a foundation for smart contracts, token issuance, and decentralized finance (DeFi), in addition to being a currency, this implies that it is more sensitive to changes in regulatory sentiment. Regulation may be seen by investors as a direct danger to these features. In contrast, Bitcoin appears to withstand regulatory shocks with reasonable stability and serves a more limited purpose as a digital commodity or store of wealth.

Comparative Analysis by Regulatory Announcement Type

We then looked at the average CAR for each asset based on six predefined regulatory categories shown in Table 1. This analysis seeks to understand which types of announcements cause the most significant market reaction.

Table 1: *Mean CAR for Bitcoin and Ethereum per Regulatory Announcement Type*

Regulatory Announcement Type	Mean CAR (BTC)	Mean CAR (ETH)
KYC/AML Regulations	-0.0040	0.0481
CBDC Initiatives	-0.0036	0.0373
Taxation Policies	-0.0066	0.0266
Bans	-0.0027	-0.0367
Securities Classifications	-0.0032	-0.0449
Other	0.0013	-0.0758

Ethereum responds positively to formalization rules like KYC/AML and CBDC Initiatives, however BTC and ETH react negatively to enforcement measures such as Bans and Securities Classifications.

```

Mean CAR by Regulatory Announcement Type (BTC):
  label
Other          0.001255
Bans           -0.002705
Securities Classifications -0.003246
CBDC Initiative -0.003550
KYC/AML        -0.003978
Taxation Policies -0.006558
Name: CAR, dtype: float64

Mean CAR by Regulatory Announcement Type (ETH):
  label
KYC/AML          0.048138
CBDC Initiative  0.037251
Taxation Policies 0.026580
Bans             -0.036694
Securities Classifications -0.044929
Other            -0.075842
Name: CAR, dtype: float64

```

Figure 5: *Mean CAR by Regulatory Announcement Type*

Figure 5 reflects the mean CAR for each regulatory announcement types as follows:

1. KYC/AML Regulations

Ethereum reacts favorably (0.0481) to KYC/AML regulation notifications, whilst Bitcoin exhibits a slight negative average response. This implies that ETH investors view KYC/AML laws as formalizing or legitimizing, especially for the Ethereum-based DeFi ecosystem, which has come under fire for privacy and legal concerns. The modestly unfavorable reaction to Bitcoin might be the result of investors that value decentralization and anonymity.

2. CBDC Initiatives

Announcements pertaining to CBDC also cause Ethereum to react favorably, although Bitcoin continues to react somewhat negatively. Ethereum's success might be related to its involvement in infrastructural debates on digital currencies issued by central banks, where

governments occasionally suggest blockchain-based solutions. However, Bitcoin may be seen as an outsider or rival to state-backed digital currencies, which might cause slight drops.

3. Taxation Policies

Announcements of taxes cause Bitcoin to moderately decrease while Ethereum sees minor increases. The disparity might be a reflection of how Ethereum users, particularly DApps, NFTs, and token issuers, prefer more transparent taxation regulations as a means of achieving widespread adoption. Bitcoin owners may be more concerned about reporting requirements and compliance constraints, especially those who are long-term or privacy-conscious.

4. Bans

For both cryptocurrencies, bans result in negative mean CARs; however, Ethereum's decline is significantly more pronounced. This demonstrated how disruptive probationary measures are for Ethereum, most likely because of its ecosystem dependencies; the ETH network is more obviously harmed if DApps or token issuers are prohibited or restricted. Bitcoin is less susceptible to prohibitions that target services or applications because it is a peer-to-peer money.

5. Securities Classifications

Ethereum is far more adversely impacted in this category as well. The market's apprehension regarding ETH's legal classification is reflected in the -0.0449 mean CAR, particularly in the light of the continuous discussions regarding whether ETH or related tokens ought to be governed as securities. Bitcoin responds more subtly to securities rules since it is less risky and is widely recognized as a commodity in major jurisdictions like the U.S.

6. Other Announcements

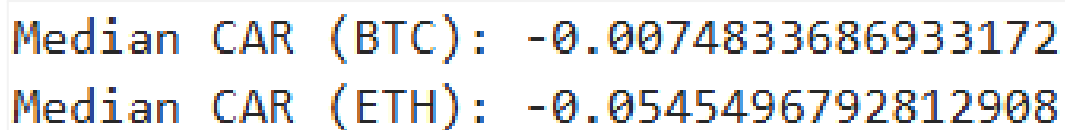
The highest negative mean CAR for Ethereum (-0.0758) is produced by this catch-all category, suggesting that news that are unclear, unclassified, or various could cause more ETH market trepidation or anxiety. The slight increase in Bitcoin's value here may indicate that investors are not particularly alarmed by the ambiguous or indirect regulatory announcements, or that they are even dismissed as unimportant or speculative chatter.

Distributional Characteristics of CARs

The median CAR is a reliable metric of central tendency and can reveal skewness in data distributions.

- **BTC:** Median CAR = **-0.0075**
- **ETH:** Median CAR = **-0.0545**

Ethereum's much lower median compared to its mean indicates a right-skewed distribution with very large positive outliers.



Median CAR (BTC): -0.0074833686933172
Median CAR (ETH): -0.0545496792812908

Figure 6: *Median CAR*

Figure 6 shows that median CARs for both assets are negative, but Etehereum's value is once more extreme. More than half of regulatory notifications result in noticeably negative returns, much more so than the norm, according to ETH's median of -0.0545.

A right-skewed distribution is suggested by the fact that ETH's median is significantly lower than its mean; this means while the majority of reactions are highly negative, a few significant positive outliers elevate the average. A more balanced reaction pattern is implied by Bitcoin's reduced mean-to-median difference, which may point to increased price stability and investor trust in the asset in spite of regulatory challenges.

Volatility in Regulatory Response

The standard deviation of CARs measures the level of dispersion or variability in market reactions.

- **BTC:** Standard Deviation = **0.0310**
- **ETH:** Standard deviation = **0.1100**

As presented in Figure 7, Ethereum has a substantially greater standard deviation, indicating a more irregular and extreme price changes in response to regulation.

Standard Deviation of CAR (BTC): 0.03104023279752792
 Standard Deviation of CAR (ETH): 0.10997215932092504

Figure 7: *Standard Deviation of CAR*

Ethereum's price responses to legislation are extremely volatile, as evidenced by its CARs, which have a standard deviation that is more than three times that of Bitcoin. This result is consistent with other findings that Ethereum is more vulnerable to legal categories (like securities law), which can have a wide range of repercussions.

In comparison, Bitcoin has significantly less fluctuation in abnormal returns, indicating a more consistent response to regulatory updates. For investors and policymakers, this distinction is critical: Ethereum's market reactions are less predictable and less stable, particularly in the short term.

Extremes in Market Reactions

Analyzing the least and highest CAR values provides the most extreme negative and positive reactions in the sample.

- **BTC:**
 - Minimum CAR = **-0.1028**
 - Maximum CAR = **0.1146**
- **ETH:**

- Minimum CAR = **-0.2741**
- Maximum CAR = **0.2097**

Ethereum once again exhibits more extreme responses on both ends, indicating higher risk and reward exposure to regulatory events/developments.

```
Minimum and Maximum CAR (BTC): -0.1028399225198933 / 0.1145874118275062
Minimum and Maximum CAR (ETH): -0.2740835290002161 / 0.2096652501927496
```

Figure 8: *Minimum and Maximum CAR*

Once more, Ethereum stands out due to much more severe market reactions, both in a positive and negative way. Figure 8 shows that the worst-case event for Bitcoin (-0.1028) is over three times deeper than the most negative ETH CAR (-0.2741), while the highest positive reaction (0.2097) is also higher than the maximum for Bitcoin (0.1146).

This broad range demonstrated that, in the context of regulatory change, Ethereum is a high-risk, high-reward asset. When faced with enforcement-heavy measures like bands or securities classifications, it can see significant sell-offs, but it can also rise quickly in response to favorable laws (such as formalization via KYC/AML or CBDC initiatives).

Because of its longer history, decentralized structure, and limited functional exposure to sectors that are subject to strict regulatory oversight, Bitcoin's smaller range supports its reputation as a more established and robust asset.

Range

The range of Cumulative Abnormal Returns (CAR) measures the spread between the maximum and minimum values, providing a simple indicator of the extent of market reactions to regulatory announcements.

BTC CAR Range: 0.2174
ETH CAR Range: 0.4838

Figure 9: *CAR Range*

The range analysis shown in Figure 9 reveals that Ethereum exhibits a significantly wider range (0.4838) compared to Bitcoin (0.2174). This indicates that Ethereum experiences more extreme market reactions to regulatory announcements, with price movements that are more than twice as volatile as Bitcoin's. The substantial difference in ranges suggests that Ethereum's market value is more sensitive to regulatory developments, potentially due to its broader functionality beyond currency (smart contracts, DeFi applications, etc.) which exposes it to a wider spectrum of regulatory concerns and interpretations.

Skewness

Skewness measures the asymmetry of the CAR distribution around its mean, highlighting whether extreme positive or negative reactions dominate.

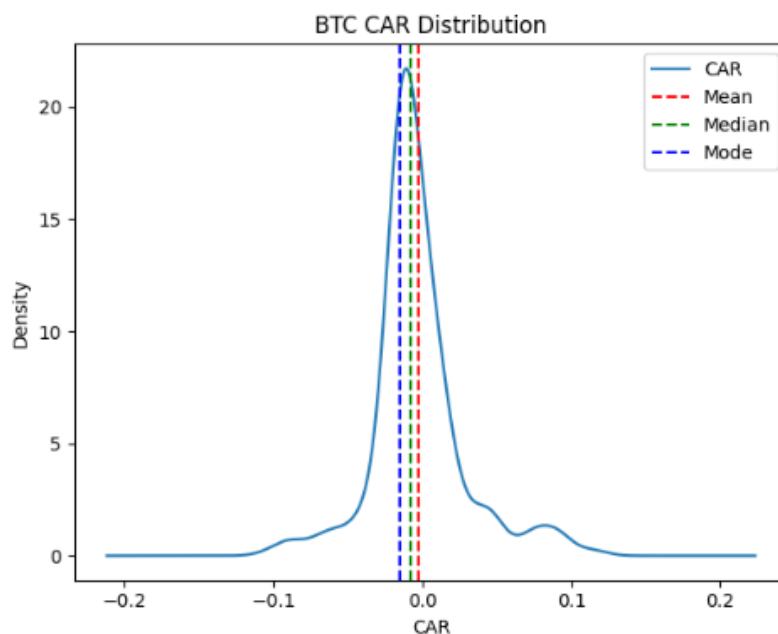


Figure 10: *Skewness of CAR for BTC*

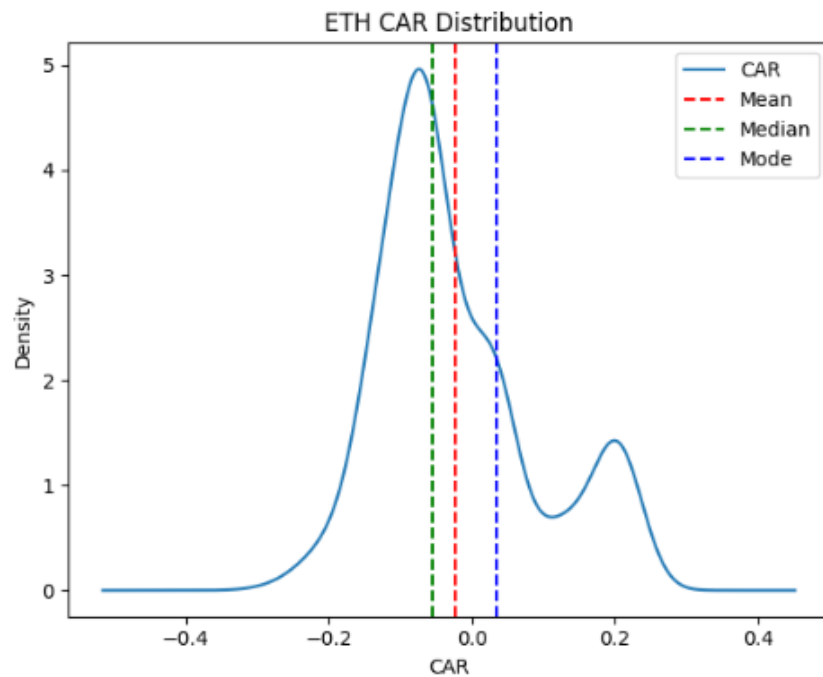


Figure 11: *Skewness of CAR for ETH*

Figures 10 and 11, respectively, shows that Bitcoin and Ethereum exhibit positive skewness—BTC at +0.80 and ETH at +0.80—indicating that, although most regulatory announcements produce returns clustered near their respective means, there are occasional large positive outliers that stretch the right tail of each distribution. Table 2 explains in practical terms, that while negative or muted reactions are common immediately following policy news, both assets can experience substantial upside surprises. For Ethereum, this pattern dovetails with its heightened sensitivity to formalization events (e.g. KYC/AML or CBDC announcements), which sometimes trigger outsized positive rebounds even as the median CAR remains negative.

Table 2: *BTC/ETH Skewness Result*

Cryptocurrency	Skewness (g ₁)	Direction	Practical meaning

Bitcoin (BTC)	+0.80	Moderately right-skewed	Rare but sizeable positive CAR spikes extend the right tail.
Ethereum (ETH)	+0.80	Moderately right-skewed	Exhibits a tail of equally large positive outliers.

Kurtosis

Kurtosis measures the "tailedness" of the probability distribution of CARs. A high kurtosis value indicates a distribution with heavy tails and a sharper peak, suggesting more frequent extreme values compared to a normal distribution.

Bitcoin (BTC) CAR distribution, Fisher-kurtosis ≈ 3.07 Ethereum (ETH) CAR distribution, Fisher-kurtosis ≈ -0.10

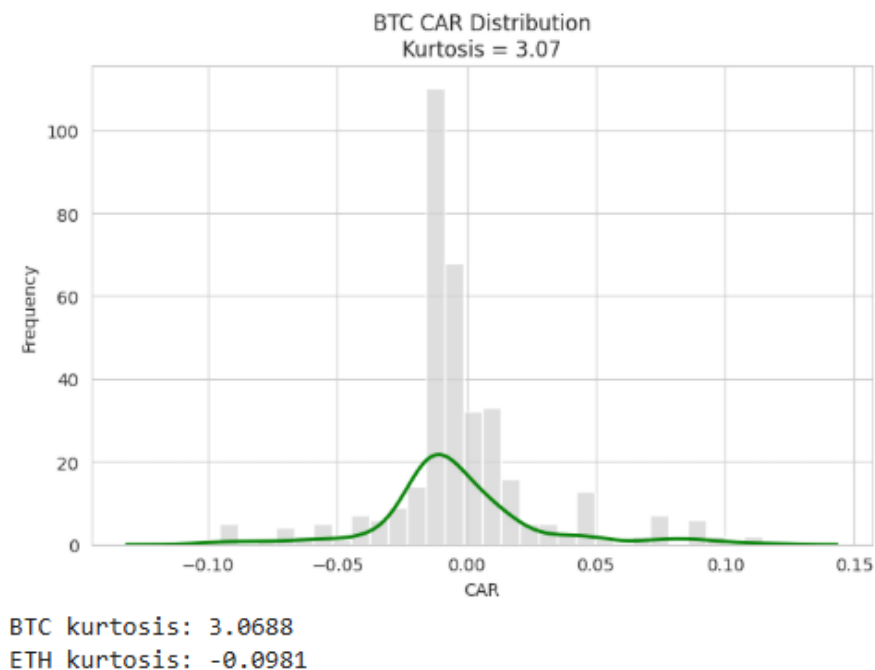


Figure 12: *Kurtosis of CAR for ETH*

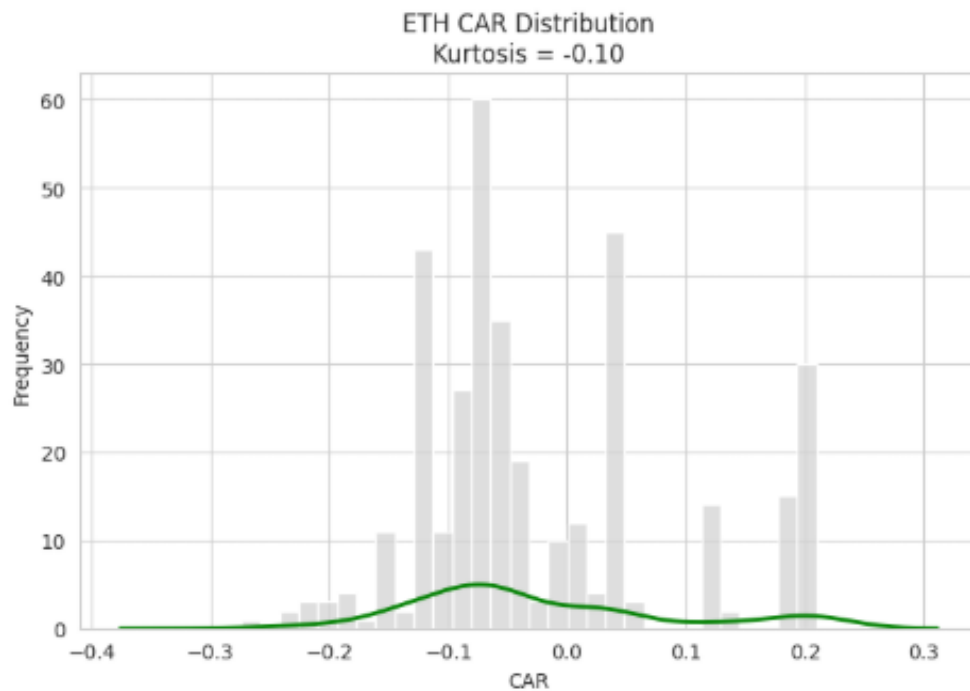


Figure 13: *Kurtosis of CAR for ETH*

The kurtosis analysis shown in Figures 12 and 13 reveals pronounced differences in the shape of CAR distributions for Bitcoin and Ethereum. Bitcoin posts a strongly positive kurtosis of +3.07, signifying a clearly leptokurtic profile: the distribution is sharply peaked at the centre while its tails are much heavier than those of a normal curve. Practically, this means most regulatory-day returns cluster tightly near the mean, but the market occasionally delivers extreme positive or negative jumps that lie far outside the typical range.

Ethereum, by contrast, records a near-zero kurtosis of -0.10, producing a distribution that is effectively mesokurtic (close to normal) and only slightly flatter than the Gaussian benchmark. In other words, ETH price reactions are more evenly dispersed across the spectrum of possible values; tail events are neither unusually frequent nor unusually scarce.

These contrasting tail behaviours reinforce earlier findings: Bitcoin’s market displays a “calm-until-it-isn’t” pattern—stable most of the time but prone to rare, outsized moves when regulation surprises. Ethereum, benefiting from a broader usage base and diverse regulatory touchpoints, instead shows a wider everyday spread (large range) without an

excess of extreme outliers. Portfolio risk models should therefore accommodate fat-tail jump risk for BTC while treating ETH returns as closer to normally distributed within the one-day event window studied.

Positive vs. Negative Count Basic Sentiment Breakdown

This subsection presents a sentiment classification of cumulative abnormal returns (CAR) in response to regulatory announcements. By distinguishing between positive and negative CAR values, it offers a general overview of whether such events tend to elicit favorable or adverse market reactions.

```

=== Positive vs. Negative CAR Count ===
CAR_sentiment
Negative      233
Positive      127

```

Figure 14: *Positive vs. Negative CAR Count for BTC*

```

=== Positive vs. Negative CAR Count ===
CAR_sentiment
Negative      235
Positive      125

```

Figure 15: *Positive vs. Negative CAR Count for ETH*

An initial examination of market reactions to regulatory announcements suggests a largely negative sentiment for Bitcoin (BTC) and Ethereum (ETH). Figure 14 shows that of all observed events, BTC documented 233 negative cumulative abnormal returns (CAR) vs only 127 positive ones. Similarly, Figure 15 illustrates ETH had 235 negative CAR events and only 125 positive. This continuous imbalance indicates that the market interprets most regulatory developments as adverse signals regardless of jurisdiction or category. The data suggest a broad short-term negative market sentiment in response to regulatory news,

emphasizing the sensitivity of both cryptocurrencies to policy changes and legal interventions.

Histogram of CAR

This subsection illustrates the frequency distribution of CAR values. This visualization aids in identifying the overall shape of the distribution, thereby providing insights into the consistency and extremity of market responses.

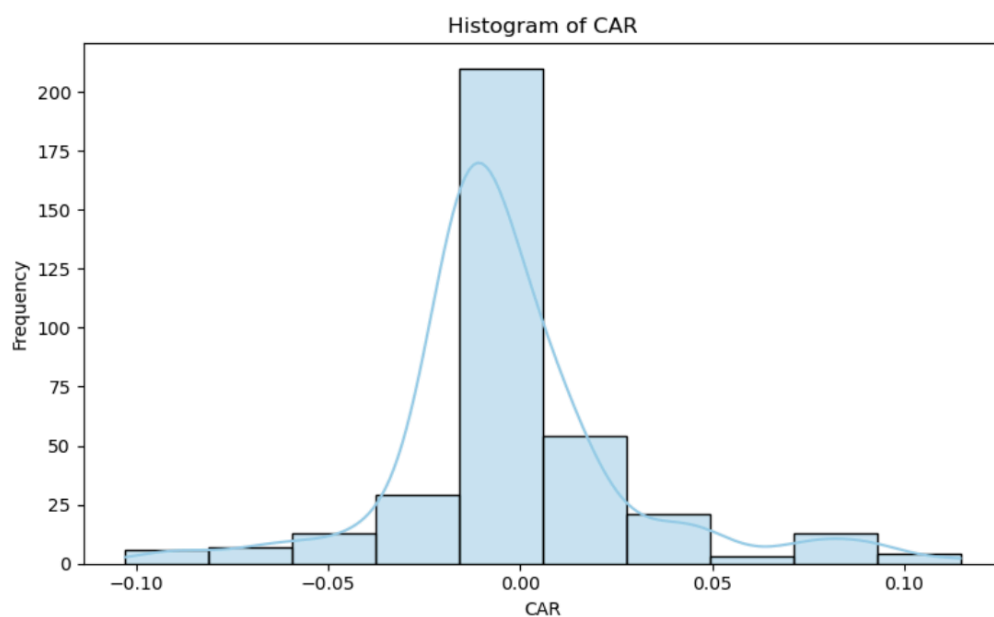


Figure 16: *Histogram of CAR for BTC*

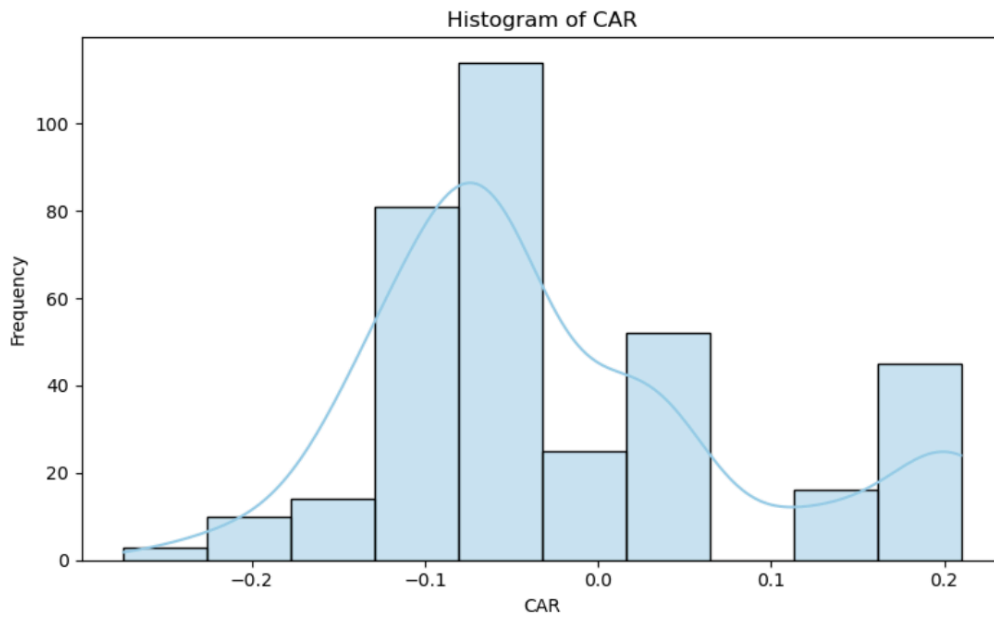


Figure 17: *Histogram of CAR for ETH*

The CAR histograms for BTC and ETH show significant differences in how each asset responds to regulatory announcements. Figure 16 shows BTC's CAR distribution appears to be approximately normal, centered around zero, with a slight positive skew. This suggests that Bitcoin often absorbs regulatory news with relatively minor price fluctuations, implying a more mature or resilient market response. In contrast, Figure 17 shows Ethereum's CAR distribution is larger, more distributed, and negatively skewed. The frequency of large negative CARs in ETH's histogram indicates that Ethereum's reactions to regulatory announcements are more pronounced and volatile. These findings suggest that ETH is more sensitive to regulatory developments and, hence, more vulnerable to adverse investor sentiment in the short term.

CAR by Regulatory Category (Label)

This subsection analyzes CAR values based on the type of regulatory announcements. By grouping events according to their regulatory label (*e.g., KYC/AML, Bans, Securities Classifications*), it facilitates the assessment of which categories tend to exert stronger market influence, either positively or negatively.

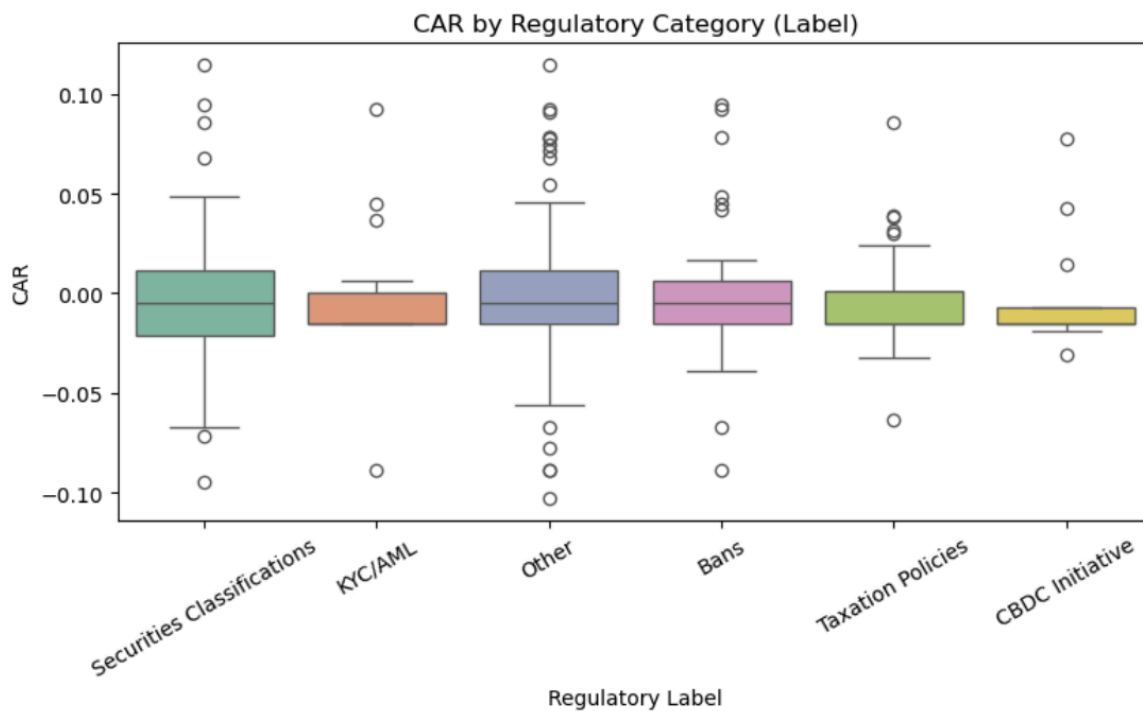


Figure 18: *CAR by Regulatory Category (Label) for BTC*

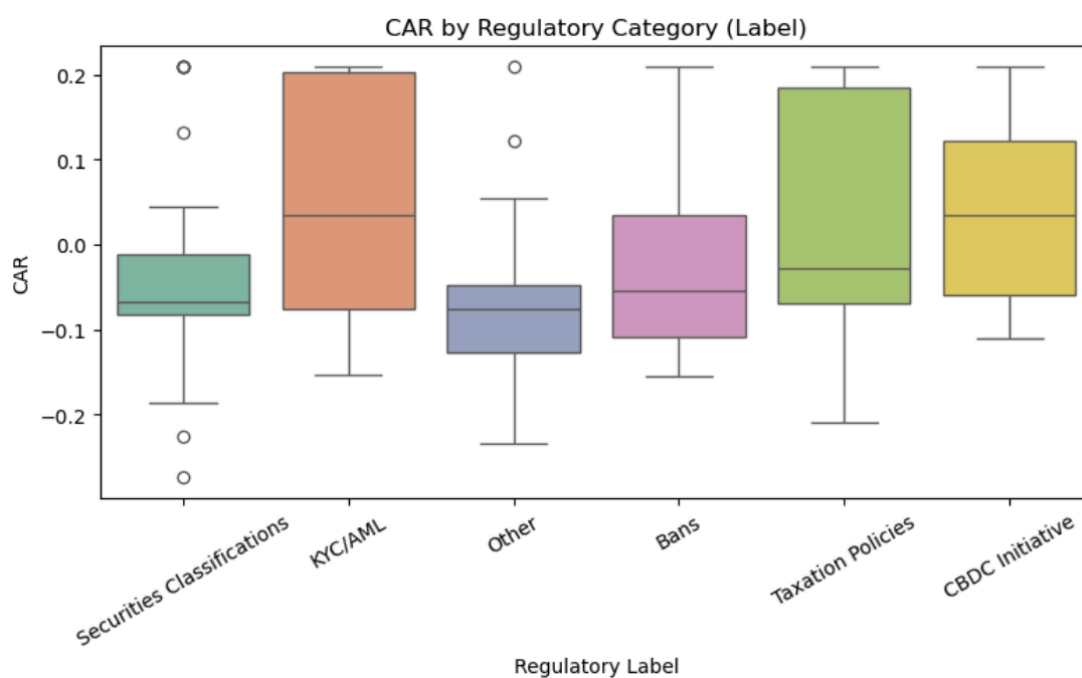


Figure 19: *CAR by Regulatory Category (Label) for ETH*

Grouping CAR by regulatory category (e.g., Bans, Taxation, Securities Classification)

Figure 18 reveals that BTC's returns remain reasonably consistent across different announcement types. The median CARs for most categories are close to zero, with very

minor variability, indicating a consistent response. On the other hand, Figure 19 illustrates that ETH exhibits stronger and more variable responses. Categories such as "KYC/AML" and "Taxation Policies" have higher median CARs and larger interquartile ranges, but "Securities Classification" generates more negative responses. The existence of outliers in ETH responses, both positive and negative, suggests that Ethereum's market reacts more strongly to specific regulatory narratives, particularly those that directly affect compliance obligations and asset classification.

Timeline Visualization Plots CARs

This subsection visualizes CAR trends over time. It enables the detection of temporal patterns, clusters of impactful events, or periods of increased market sensitivity, providing context for how regulatory reactions may evolve in different market conditions.

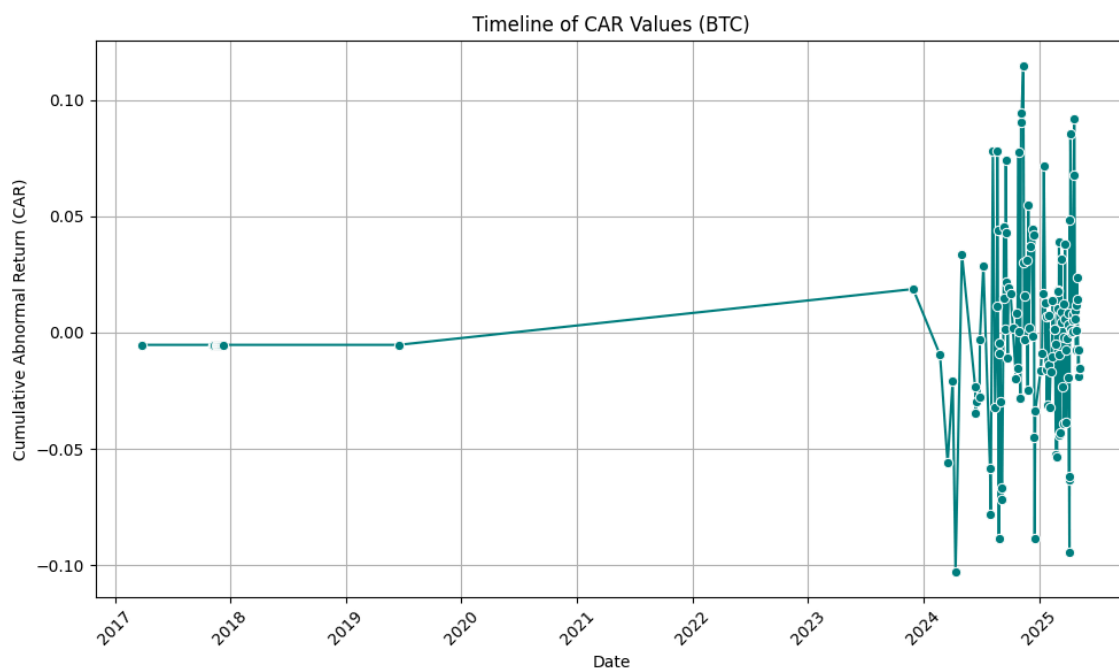


Figure 20: *Timeline of CAR Values for BTC*

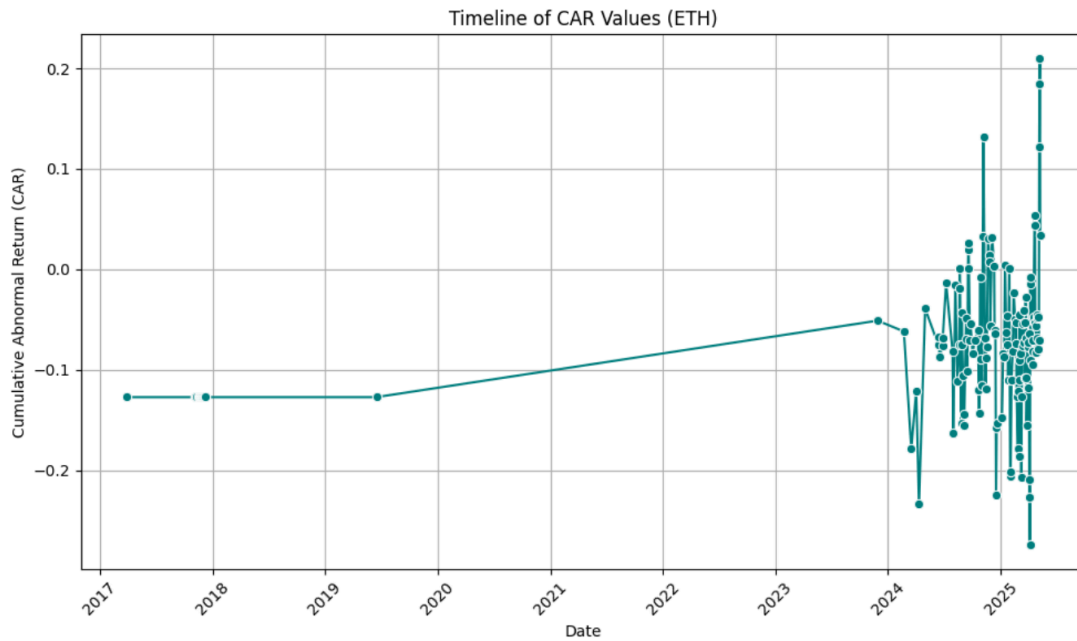


Figure 21: *Timeline of CAR Values for ETH*

The timeline visualization of CARs for BTC and ETH reveals considerable changes in market responsiveness to regulatory events throughout time. From 2017 to 2022, Figure 20 shows that the BTC's CAR values remained largely neutral and stable, with a notable increase in volatility and clustering of event-driven movements beginning in 2023. This indicates that the BTC market has been more sensitive to regulatory announcements in recent years. In contrast, Figure 21 shows that the ETH has a greater negative baseline CAR in older years, particularly before 2023, indicating a historically more negative market reaction to regulatory news. After 2023, ETH's CARs show increasing dispersion and volatility, with larger variations than BTC, implying that Ethereum may be more reactive and, therefore, more vulnerable to regulatory developments. Over time, the divergence in CAR behavior between BTC and ETH highlights possible differences in investor sentiment, market maturity, and perceived regulatory risk between the two cryptocurrencies.

Outlier Detection

Outlier detection shown in Figures 22 and 23, identifies regulatory events that resulted in unusually large market reactions, providing insights into which specific types of announcements have the greatest impact on cryptocurrency prices.

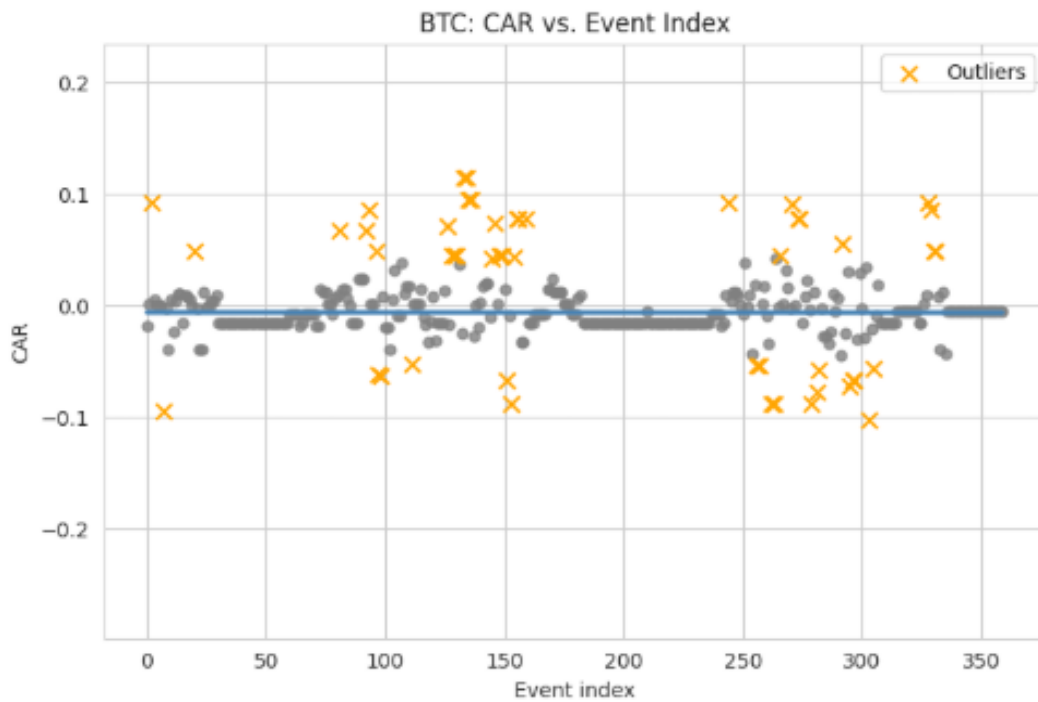


Figure 22: *BTC-CAR vs Event Index*

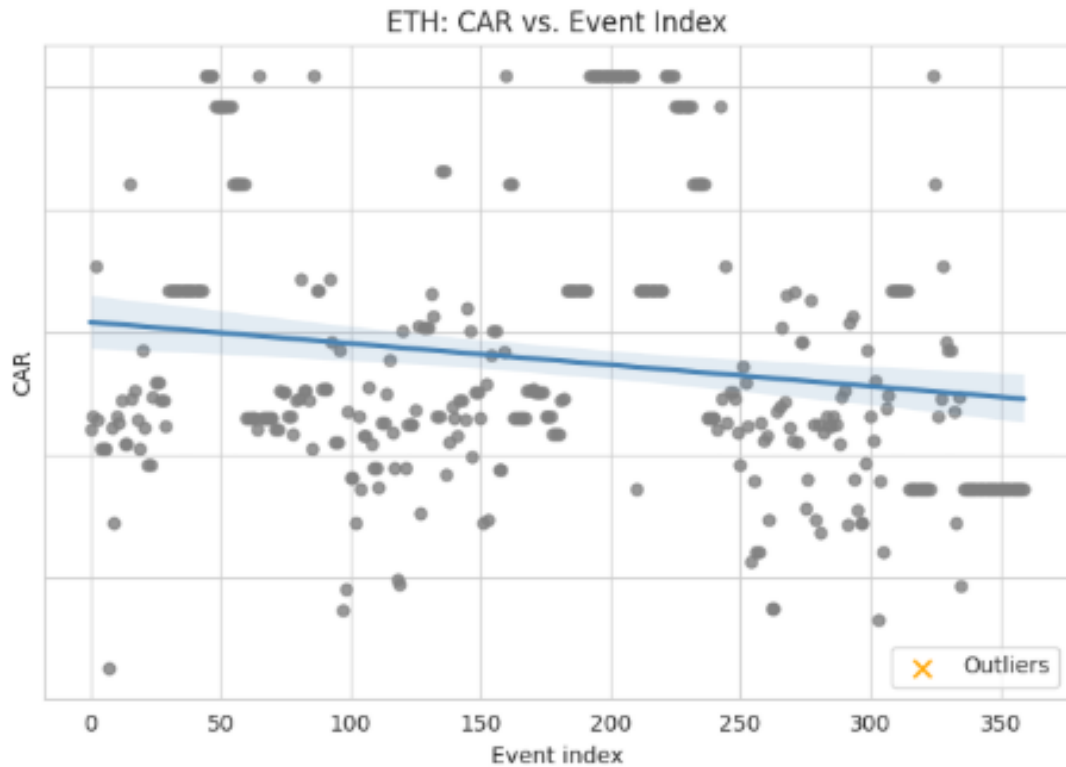


Figure 23: *ETH-CAR vs Event Index*

Table 3 shows usage of the Interquartile Range (IQR) method to identify outliers highlights distinct patterns in extreme market responses for Bitcoin and Ethereum; Bitcoin displays 50 outlier events, representing approximately 14.1% of all observations, with a relatively narrow range between -0.0501 and 0.0424. This suggests that while the majority of Bitcoin's market reactions to regulatory announcements remain modest and clustered around the mean, certain events trigger significant, exceptional price movements.

Table 3: *BTC/ETH Outlier Detection Result*

Asset	IQR lower bound	IQR upper bound	Outlier count
BTC	-0.0501	+0.0424	50
ETH	-0.2760	+0.2198	0

In contrast, Ethereum exhibits no statistical outliers within its considerably wider range of -0.2760 to 0.2198. Despite its broader daily volatility, Ethereum's CAR values remain consistently within these boundaries, indicating that its substantial price reactions are more evenly distributed and do not typically reach statistical extremes defined by the IQR method.

A deeper review of Bitcoin's outlier events reveals that its extreme price responses predominantly occur around major regulatory announcements related to securities classifications and outright bans, particularly from jurisdictions such as the United States and China. This aligns with Bitcoin's narrative as a focused digital asset whose valuation is particularly sensitive to existential regulatory threats.

Ethereum's absence of IQR-defined outliers suggests its market responses are more broadly spread across regulatory events without reaching extreme statistical thresholds. This consistent response pattern reflects Ethereum's diverse functional applications and broader ecosystem, which absorbs regulatory shocks in a more proportional manner without singularly dominant price spikes.

These findings underline important distinctions in investor perceptions and reactions: Bitcoin investors react dramatically to existential regulatory threats, resulting in occasional severe outliers, whereas Ethereum's market exhibits consistent and proportional sensitivity across a wider spectrum of regulatory developments.

Mann–Whitney U Test

The Mann–Whitney U Test was conducted to compare the cumulative abnormal returns (CARs) of Bitcoin (BTC) and Ethereum (ETH), assessing whether the market reactions to regulatory announcements significantly differ between these two cryptocurrencies.

$$\mathbf{U\text{-}statistic = 84\ 505}$$

$$\mathbf{p\text{-value}} = 1.52 \times 10^{-12}$$

The Mann–Whitney U Test yielded a highly significant p-value ($p < 0.001$), indicating strong statistical evidence against the null hypothesis that Bitcoin and Ethereum CAR distributions are identical. The substantial difference in distributions reflects that Ethereum's and Bitcoin's market reactions to regulatory news are fundamentally distinct.

Bitcoin's narrower range of reactions, highlighted by a tight distribution around the mean and periodic extreme outliers, contrasts with Ethereum's broader yet consistently negative-leaning response. This aligns with our earlier findings where Bitcoin reacts strongly to major existential threats (such as outright bans), while Ethereum displays broader sensitivity to a wide array of regulatory changes.

The statistical difference detected by the Mann–Whitney U Test reinforces the conclusion that regulatory developments have asset-specific impacts, driven by their differing market structures, use cases, and investor bases. Hence, regulatory risk assessment and management strategies should explicitly differentiate between these two leading cryptocurrencies.

One-Sample t-Test

The One-Sample t-Test was performed to determine if the average cumulative abnormal returns (CARs) for Bitcoin and Ethereum significantly deviate from zero following regulatory announcements, with the computed results summarized in Table 4.

Table 4: *BTC/ETH One-Sample t-Test Result*

Cryptocurrency	t-statistic	p-value	Significance Level
Bitcoin (BTC)	-1.7684	0.0778	Marginally significant ($p > 0.05$)

Ethereum (ETH)	-3.9438	0.0001	Highly significant ($p < 0.001$)
----------------	---------	--------	------------------------------------

Bitcoin (BTC): The t-statistic of -1.7684 with a p-value of 0.0778 indicates that Bitcoin's average CAR is marginally negative but not statistically significant at the standard 5% level. This suggests that regulatory announcements, on average, cause minor negative reactions in Bitcoin's price, though this effect is not robust enough to rule out random chance.

Ethereum (ETH): In contrast, Ethereum shows a strongly significant negative average CAR, with a t-statistic of -3.9438 and a very low p-value (0.0001). This confirms that Ethereum consistently experiences meaningful negative market reactions in the wake of regulatory announcements, significantly different from zero.

These findings underline a fundamental difference in investor sentiment and price sensitivity between the two cryptocurrencies. Ethereum's significant negative reaction aligns with its broader exposure to regulatory risks due to its extensive application spectrum. Bitcoin's reaction, while negative on average, appears relatively muted, reinforcing its established narrative as a relatively resilient digital asset, particularly outside of extreme regulatory circumstances.

Impact of Regulatory Announcements By Jurisdiction

Mean & Median CAR by Country

This subsection analyzes the average and typical short-term market response of Bitcoin (BTC) and Ethereum (ETH) to regulatory announcements originating from three major jurisdictions: the United States, China, and the European Union. By examining both the **mean** and **median** cumulative abnormal returns (CAR), the study assesses not only the direction of market reactions but also whether the effects are skewed or consistent across events within each region.

Table 5: BTC/ETH Mean and Median CAR by Country

Cryptocurrency	Country	Mean	Median
Bitcoin (BTC)	China	-0.005812	-0.005255
Bitcoin (BTC)	European Union	0.008015	0.001233
Bitcoin (BTC)	United States	-0.005187	-0.015403
Ethereum (ETH)	China	-0.087492	-0.090075
Ethereum (ETH)	European Union	-0.036878	-0.053248
Ethereum (ETH)	United States	0.012392	-0.045433

The summarized results in Table 5 present the mean and median cumulative abnormal returns (CAR) for Bitcoin (BTC) and Ethereum (ETH) in response to regulatory announcements across three key jurisdictions: China, the European Union, and the United States. These values provide a comparative view of both the magnitude and consistency of market reactions by country.

For Bitcoin (BTC), announcements from China led to a slightly negative average CAR (-0.005812) with a closely aligned median (-0.005255), indicating consistently mild but negative short-term effects. In contrast, European Union announcements produced a positive mean CAR of 0.008015 , although the median was lower (0.001233), suggesting that a few strong positive events may have skewed the average. The United States also exhibited a negative average CAR (-0.005187), but with a more negative median (-0.015403), implying

that the typical BTC response to U.S. regulatory actions was more strongly adverse than the mean alone would suggest.

For Ethereum (ETH), reactions were more pronounced. Regulatory announcements from China triggered the most negative impact, with a mean CAR of -0.087492 and a median of -0.090075 , reflecting a clear and consistent adverse market response. The European Union also had a negative effect on ETH (mean: -0.036878 ; median: -0.053248), though less severe than China's. U.S. regulations yielded a slightly positive mean CAR (0.012392) but a notably negative median (-0.045433), indicating highly variable outcomes—some events drove price increases, while the typical response remained negative.

Overall, the results indicate that China's announcements consistently triggered negative market reactions, especially for ETH. The U.S. responses were mixed, with ETH showing higher volatility than BTC, while the EU's effects were comparatively moderate, particularly for BTC. These findings suggest that market sensitivity is not only asset-specific but also highly dependent on the regulatory style and market perception of the originating jurisdiction.

Jurisdiction-by-Asset Comparison

This subsection presents the distribution of CAR values across different countries. The comparison highlights regional differences in how markets respond to regulatory announcements.

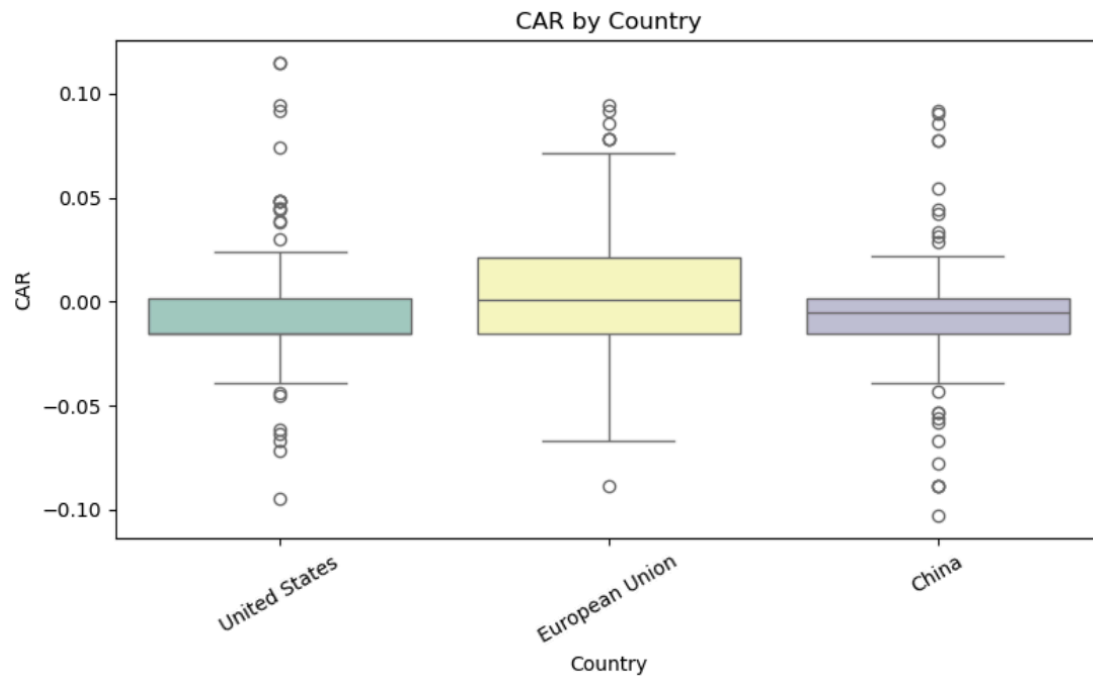


Figure 24: *CAR by Country for BTC*

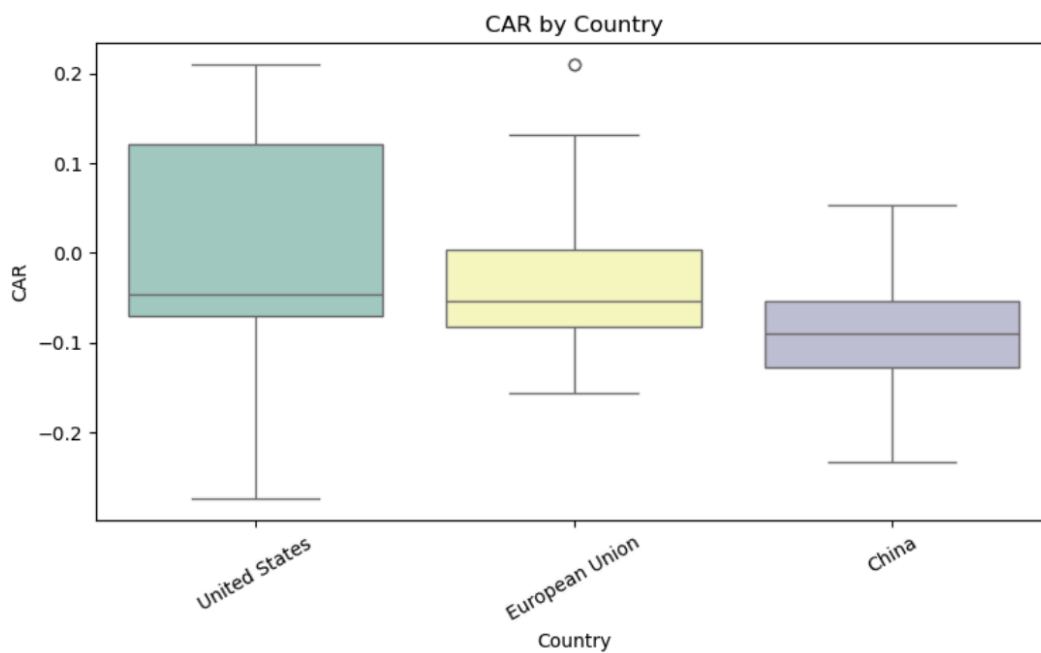


Figure 25: *CAR by Country for ETH*

Analyzing CAR by the source country of regulatory news reveals more differences between BTC and ETH. Figure 24 reveals Bitcoin's reactions have remained neutral across announcements from major jurisdictions such as the United States, China, and the European Union. While U.S. announcements cause slightly negative CARs for BTC, the impact is mild

and constant. However, Figure 25 illustrates that Ethereum shows more pronounced country-specific responses. U.S. announcements are associated with a broader range of CARs, including some strongly positive outliers, which could reflect investor confidence in more specific regulatory guidance. Conversely, news from China consistently results in negative CARs for ETH, showing a greater vulnerability to regulatory measures from that jurisdiction. The EU's influence over ETH is more balanced, with moderate reactions. These findings indicate that Ethereum's price is more sensitive to the geographic source of regulation, especially in markets with high enforcement or jurisdictional uncertainty.

Distributional Differences (Standard Deviation & Range by Country)

Standard deviation, minimum and maximum CAR values, and the range between extremes help assess the volatility and dispersion of market reactions. These metrics reveal not only the average impact but also how consistently (or inconsistently) markets respond to different countries' regulations. Table 6 summarizes these distributional differences for both Bitcoin (BTC) and Ethereum (ETH).

Table 6: BTC/ETH Standard Deviation & Range by Country

Cryptocurrency	Country	Standard Deviation	Min	Max	Range
Bitcoin (BTC)	China	0.035084	-0.005255	0.091999	0.194839
Bitcoin (BTC)	European Union	0.036988	-0.088520	0.094425	0.182944
Bitcoin (BTC)	United States	0.025752	-0.094661	0.114587	0.209249

Ethereum (ETH)	China	0.066070	-0.233693	0.053662	0.287355
Ethereum (ETH)	European Union	0.078548	-0.155848	0.209665	0.365513
Ethereum (ETH)	United States	0.120508	-0.274084	0.209665	0.483749

For Bitcoin, CAR values from China and the European Union exhibit relatively similar variability, with standard deviations of 0.035 and 0.037 respectively, and modest ranges just below 0.20. This suggests that while BTC reacts negatively to some announcements from these regions, the magnitude of reaction remains relatively constrained. Interestingly, the United States shows the lowest BTC volatility, with a standard deviation of only 0.0258, suggesting that U.S. regulations have a more stable and predictable impact on Bitcoin's short-term behavior.

In contrast, Ethereum's CAR responses are markedly more volatile across all jurisdictions. ETH events from the United States stand out with a standard deviation of 0.1205 and a maximum observed range of 0.4837 — more than double that of BTC — indicating extreme swings in short-term market sentiment. Even in China and the EU, ETH shows broader response dispersion (ranges of 0.2874 and 0.3655, respectively), which implies that Ethereum is more sensitive and less stable in its short-term response to regulatory signals, particularly from Western jurisdictions.

These findings highlight that jurisdictional origin not only affects the direction of market reaction (as seen in Section 3.5.1) but also its volatility, with Ethereum notably exhibiting higher distributional spread than Bitcoin in all regions. This increased sensitivity

has implications for investors and regulators alike, suggesting that Ethereum markets may be more prone to uncertainty or overreaction.

Conclusion

This study investigated the short-term market reactions of Bitcoin (BTC) and Ethereum (ETH) to regulatory announcements issued between 2023 and 2025, using an event study methodology. The research aimed to measure abnormal price behavior surrounding significant policy events, determine whether the impact differed across countries and regulatory categories, and compare the magnitude and direction of these effects between the two leading crypto assets.

Key findings revealed that Ethereum exhibited significantly greater short-term sensitivity to regulatory actions than Bitcoin, both in terms of average CAR values and volatility. ETH CARs were more negative overall and displayed wider ranges and higher standard deviations across all jurisdictions, indicating that Ethereum markets are more prone to volatility and abrupt sentiment shifts. BTC, by contrast, showed more contained and consistent reactions, particularly in response to U.S. and EU announcements. Ethereum's higher sensitivity may stem from structural inefficiencies and heightened behavioral reactivity, as observed in earlier studies.

Among jurisdictions, China consistently triggered the most negative and uniform responses across both assets, with a clear pattern of market aversion to restrictive policies such as outright bans and enforcement crackdowns. Events originating from the United States showed mixed results; while enforcement actions (e.g., securities designations) were associated with strongly negative CARs, more formal regulatory clarifications occasionally produced neutral or even slightly positive responses, especially for BTC. The European Union stood out as the most balanced and stable region, with CARs that were generally smaller in magnitude and less volatile, reflecting a more procedural and gradual regulatory

environment. Differences in jurisdictional approach explain variation in market reactions, consistent with the global fragmentation documented in prior regulatory research.

Categorically, Securities Classifications and Bans had the strongest negative effects on market behavior, confirming that announcements which threaten access, legitimacy, or asset status are interpreted by investors as high-risk. On the other hand, KYC/AML Regulations and CBDC Initiatives were often associated with neutral or even modestly positive reactions, especially for Ethereum. This suggests that markets distinguish between hostile or disruptive measures and those perceived as formalization or infrastructure-building efforts. It also aligns with previous literature confirming the market's sensitivity to regulatory interventions, particularly enforcement-heavy actions.

Statistical tests such as the one-sample t-test confirmed that many of the observed CARs were significantly different from zero, validating that short-term market behavior is measurably influenced by regulatory announcements. Comparative tests across categories and jurisdictions using Kruskal–Wallis and Mann–Whitney U methods further substantiated that the impact of regulation is not uniform, but rather highly dependent on both what is announced and where it is announced.

In conclusion, this study underscores that regulatory announcements are powerful market-moving events, with Ethereum being more sensitive than Bitcoin, and China's actions constantly prompting the most negative reactions.

These findings carry implications for both investors and policymakers. Traders and institutions can benefit from closer monitoring of jurisdiction-specific announcements to anticipate volatility, while regulators may consider the timing, framing, and transparency of policy communication to mitigate disruptive market impacts.

These findings carry important implications for a wide range of stakeholders within the cryptocurrency ecosystem. For investors and institutional traders, understanding the jurisdictional origin and regulatory category of announcements offers a strategic advantage. The study showed that Chinese bans and securities-related designations from the U.S. often trigger immediate negative market reactions—especially for Ethereum—while announcements categorized under KYC/AML or CBDC initiatives generally produce more stable or even favorable outcomes. Investors can use this insight to calibrate short-term trading strategies, apply protective hedging ahead of major policy releases, or even construct risk models that account for the type and origin of regulatory news as predictive variables in volatility forecasting.

Policymakers and regulators can also draw important lessons. The data indicates that enforcement-heavy, unpredictable regulatory actions—particularly from China and the U.S.—tend to trigger greater market disruption, while more structured and transparent communications, as seen in the EU, result in less volatile reactions. This suggests that governments aiming to regulate the crypto space without inciting instability should prioritize clarity, predictability, and stakeholder engagement in their regulatory rollout. Thoughtful timing, detailed guidance, and pre-announcement consultations could help soften adverse market effects and reduce investor panic. Moreover, distinguishing between consumer protection and outright prohibition may improve market resilience and preserve innovation in the space.

Financial analysts and economists will also find the results of this study particularly valuable. By quantifying the short-term impact of regulatory announcements on cryptocurrency markets, the research offers a framework for more precise modeling of market sentiment, investor behavior, and price volatility. The ability to distinguish between

asset-specific and jurisdiction-specific effects enhances forecasting accuracy, especially in periods of regulatory uncertainty. These insights can be integrated into macroeconomic analyses, digital asset valuation models, and market outlook reports, enabling analysts to contextualize price movements within a broader policy environment.

Finally, data scientists and analysts in the crypto space can leverage these findings to develop machine learning models that incorporate real-time regulatory inputs into asset pricing, trading signals, or portfolio stress testing. By quantifying the short-term impact of different regulatory signals, this study establishes a practical foundation for building more responsive, risk-aware financial tools in an increasingly complex policy landscape.

Recommendations

To build on the findings of this study, future research should consider expanding the event window beyond the current $[-1, 0, +1]$ range. A longer window, such as $[-3, +3]$ or $[-5, +5]$, may capture delayed market reactions, pre-event speculation, and extended volatility effects. Additionally, using dynamic models to estimate expected returns—like multi-day averages or CAPM-based benchmarks—could improve the accuracy of abnormal return calculations.

Researchers are also encouraged to incorporate broader market indicators or sentiment variables to better isolate regulatory effects. Expanding the scope to include more jurisdictions, such as South Korea, Singapore, or the UAE, would enhance the global relevance of the findings. These improvements can lead to a more comprehensive understanding of how different types of regulations influence crypto markets over time.

References

- Amsden, R. (2019). *Event Studies with Crypto Asset Returns*.
https://spectrum.library.concordia.ca/id/eprint/985163/1/Amsden_MSc_S2019.pdf
- Auer, R., & Claessens, S. (2018). Regulating cryptocurrencies: assessing market reactions.
Www.bis.org. https://www.bis.org/publ/qtrpdf/r_qt1809f.htm
- Bains, P., Ismail, A., Melo, F., & Sugimoto, N. (2022). Regulating the Crypto Ecosystem The Case of Unbacked Crypto Assets. In *IMF*. IMF.
<https://www.imf.org/-/media/Files/Publications/FTN063/2022/English/FTNEA2022007.ashx>
- Bank for International Settlements (BIS). (2023). III. Blueprint for the future monetary system: improving the old, enabling the new. *Www.bis.org*.
<https://www.bis.org/publ/arpdf/ar2023e3.htm>
- Baur, D. G., & Dimpfl, T. (2018, March 1). *Excess Volatility as an Impediment for a Digital Currency*. Papers.ssrn.com.
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2949754
- Bonaparte, Y., & Bernile, G. (2023). A New “Wall Street Darling?” Effects of Regulation Sentiment in Cryptocurrency Markets. *Finance Research Letters*, 103376.
<https://doi.org/10.1016/j.frl.2022.103376>
- Brini, A., & Lenz, J. (2024). A comparison of cryptocurrency volatility-benchmarking new and mature asset classes. *Financial Innovation*, 10(1).
<https://doi.org/10.1186/s40854-024-00646-y>
- Chainalysis. (2022, January 6). *Crypto Crime Trends for 2022: Illicit Transaction Activity Reaches All-Time High in Value, All-Time Low in Share of All Cryptocurrency Activity*. Chainalysis.
<https://www.chainalysis.com/blog/2022-crypto-crime-report-introduction/>

- Chokor, A., & Alfieri, E. (2021). Long and short-term impacts of regulation in the cryptocurrency market. *The Quarterly Review of Economics and Finance*, 81, 157–173. <https://doi.org/10.1016/j.qref.2021.05.005>
- Christensen, K., Timmermann, A., & Bezirgen Veliyev. (2023). Warp Speed Price Moves: Jumps after Earnings Announcements. *SSRN Electronic Journal*.
<https://doi.org/10.2139/ssrn.4422376>
- Coelho, H., Apostolides, A., Bear, K., Clark, N., Fleichman, N. C. de L., Letsiou, K., Singh, A., & Zhang, B. (2024, November). *2nd Global Cryptoasset Regulatory Landscape Study - CCAF publications - Cambridge Judge Business School*. Cambridge Judge Business School.
<https://www.jbs.cam.ac.uk/faculty-research/centres/alternative-finance/publications/2nd-global-cryptoasset-regulatory-landscape-study/>
- Corbet, S., Lucey, B., Urquhart, A., & Yarovaya, L. (2019). Cryptocurrencies as a financial asset: A systematic analysis. *International Review of Financial Analysis*, 62, 182–199.
<https://doi.org/10.1016/j.irfa.2018.09.003>
- CPI. (2024, January 2). *US Leads Global Crackdown on Crypto: 2023 Recap*. PYMNTS.com.
<https://www.pymnts.com/cpi-posts/us-leads-global-crackdown-on-crypto-2023-recap/>
- EY Global. (2023, March 23). *The outlook for global tax policy and controversy in 2023 now available!* Ey.com.
<https://globaltaxnews.ey.com/news/2023-5352-the-outlook-for-global-tax-policy-and-controversy-in-2023-now-available>
- FATF. (2021). *Updated Guidance for a Risk-Based Approach to Virtual Assets and Virtual Asset Service Providers*. Wwww.fatf-Gafi.org.

- <https://www.fatf-gafi.org/en/publications/Fatfrecommendations/Guidance-rba-virtual-assets-2021.html>
- Foley, S., Karlsen, J. R., & Putniņš, T. J. (2019). Sex, Drugs, and Bitcoin: How Much Illegal Activity Is Financed through Cryptocurrencies? *The Review of Financial Studies*, 32(5), 1798–1853.
- Frediani, M. E. A. (2024). Crafting the Future of Finance: A Comparative Analysis of Cryptocurrency Regulation in the Global Economy. *Journal of Financial Risk Management*, 13(1), 193–206. <https://doi.org/10.4236/jfrm.2024.131010>
- Hinman, W. (2018). *SEC.gov | Digital Asset Transactions: When Howey Met Gary (Plastic)*. [Www.sec.gov](http://www.sec.gov).
<https://www.sec.gov/newsroom/speeches-statements/speech-hinman-061418>
- Katsiampa, P. (2019). Volatility co-movement between Bitcoin and Ether. *Finance Research Letters*, 30. <https://doi.org/10.1016/j.frl.2018.10.005>
- Li, B., & Sugimoto, N. (2023, January 18). *Crypto Contagion Underscores Why Global Regulators Must Act Fast to Stem Risk*. IMF.
<https://www.imf.org/en/Blogs/Articles/2023/01/18/crypto-contagion-underscores-why-global-regulators-must-act-fast-to-stem-risk>
- Li, M. (2021, October 26). *The Rules of Crypto: A Comparison of U.S. and Chinese Regulatory Policies – The Network*. Berkeley.edu.
<https://sites.law.berkeley.edu/thenetwork/2021/10/26/the-rules-of-crypto-a-comparison-of-u-s-and-chinese-regulatory-policies/>
- Lyócsa, Š., Molnár, P., Plíhal, T., & Širaňová, M. (2020). Impact of macroeconomic news, regulation and hacking exchange markets on the volatility of bitcoin. *Journal of Economic Dynamics and Control*, 119(C).
<https://ideas.repec.org/a/eee/dyncon/v119y2020ics0165188920301482.html>

- Malkiel, B. G. (2003). The Efficient Market Hypothesis and Its Critics. *Journal of Economic Perspectives*, 17(1), 59–82.
- Nakagawa, K., & Sakemoto, R. (2022). Market Uncertainty and Correlation Between Bitcoin and Ether. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.4076411>
- OECD. (2020, October 12). *Taxing Virtual Currencies: An Overview of Tax Treatments and Emerging Tax Policy Issues - OECD*. www.oecd.org.
<https://www.oecd.org/tax/tax-policy/taxing-virtual-currencies-an-overview-of-tax-treatments-and-emerging-tax-policy-issues.htm>
- OSL Research Team. (2025). *OSL-Hong Kong's most trusted Crypto Trading Platform | Buy and Sell Bitcoin, Ethereum*. [Osl.com](http://osl.com).
<https://osl.com/academy/article/the-growing-need-for-clear-crypto-regulations-in-2025>
- Saggu, A., Ante, L., & Kopiec, K. (2023). *Uncertain Regulations, Definite Impacts: The Impact of the U.S. Securities and Exchange Commission's Regulatory Interventions on Crypto Assets*. [Ucsd.edu](http://ucsd.edu).
<https://www.sec.gov/files/ctf-input-arte-2025-02-19.pdf#:~:text=associated%20with%20negative%20abnormal%20returns,negative%20abnormal%20returns%20following%20SEC>
- Shanaev, S., Sharma, S., Shuraeva, A., & Ghimire, B. (2019). Taming the Blockchain Beast? Regulatory Implications for the Cryptocurrency Market. *SSRN Electronic Journal*.
<https://doi.org/10.2139/ssrn.3397939>
- Shine, F. (2022, January 31). *What's behind China's cryptocurrency ban?* World Economic Forum.
<https://www.weforum.org/stories/2022/01/what-s-behind-china-s-cryptocurrency-ban/>

Shine, I. (2024, May 2). *How are crypto regulations changing around the world?* World Economic Forum.

<https://www.weforum.org/stories/2024/05/global-cryptocurrency-regulations-changing/>

Timmermann, A. (2025). *Earnings News Cause Immediate Stock Price Jumps, Sometimes Moving Whole Market*. Ucsd.edu.

<https://today.ucsd.edu/story/earnings-news-cause-immediate-stock-price-jumps-sometimes-moving-whole-market>

U.S. SEC. (2023). *SEC.gov | Cyber, Crypto Assets and Emerging Technology*. Sec.gov.

<https://www.sec.gov/about/divisions-offices/division-enforcement/cyber-crypto-assets-emerging-technology>

U.S. Treasury. (2022). *The Future of Money and Payments Report Pursuant to Section 4(b) of Executive Order 14067*.

<https://home.treasury.gov/system/files/136/Future-of-Money-and-Payments.pdf>

World Economic Forum (WEF). (2020, October 14). *Global Standards Mapping Initiative: An overview of blockchain technical standards*. World Economic Forum.

<https://www.weforum.org/publications/global-standards-mapping-initiative-an-overview-of-blockchain-technical-standards/>

Zhang, P., Xu, K., & Qi, J. (2023). The impact of regulation on cryptocurrency market

volatility in the context of the COVID-19 pandemic - evidence from China. *Economic Analysis and Policy*, 80, 222–246. <https://doi.org/10.1016/j.eap.2023.08.015>

Appendix

Dataset link:  [KalApache-FinalProjectDataset](#)

Github link: <https://github.com/KinGabriel/Global-Regulatory-Analytics-Model.git>